

UNIT-III

RADIATION

Thermal radiation is electromagnetic radiation generated by the thermal motion of charged particles in matter. All matter with a temperature greater than absolute zero emits thermal radiation. When the temperature of the body is greater than absolute zero, inter-atomic collisions cause the kinetic energy of the atoms or molecules to change. This results in charge-acceleration and/or dipole oscillation which produces electromagnetic radiation, and the wide spectrum of radiation reflects the wide spectrum of energies and accelerations that occur even at a single temperature.

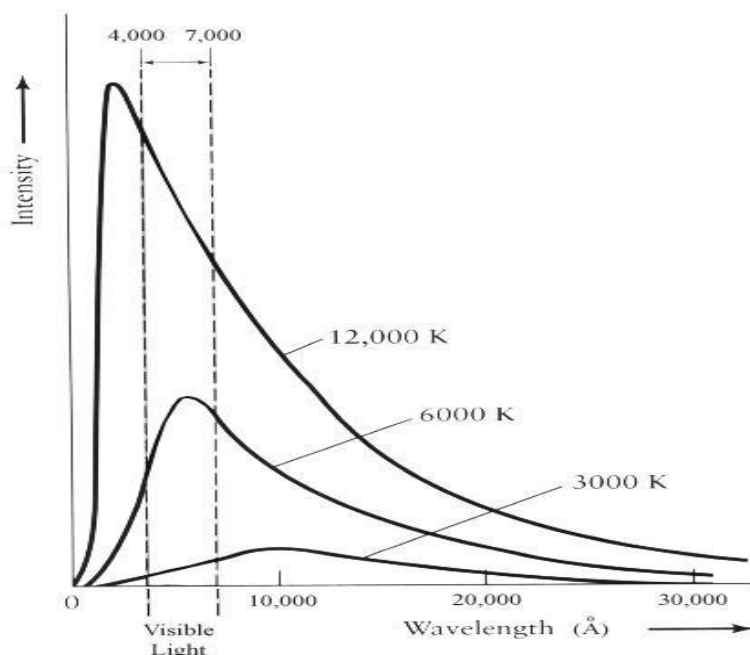
Sunlight is part of thermal radiation generated by the hot plasma of the Sun. The Earth also emits thermal radiation, but at a much lower intensity and different spectral distribution (infrared rather than visible) because it is cooler. The Earth's absorption of solar radiation, followed by its outgoing thermal radiation are the two most important processes that determine the temperature and climate of the Earth.

A **blackbody** refers to an opaque object that emits *thermal radiation*. A perfect blackbody is one that absorbs all incoming light and does not reflect any. At room temperature, such an object would appear to be perfectly black (hence the term *blackbody*). However, if heated to a high temperature, a blackbody will begin to glow with *thermal radiation*.

In fact, all objects emit thermal radiation (as long as their temperature is above Absolute Zero, or -273.15 degrees Celsius), but no object emits thermal radiation perfectly; rather, they are better at emitting/absorbing some wavelengths of light than others. These uneven efficiencies make it difficult to study the interaction of light, heat and matter using normal objects.

At the beginning of the 20th century, scientists Lord Rayleigh, and Max Planck (among others) studied the blackbody radiation using such a device. After much work, Planck was able to empirically describe the intensity of light emitted by a blackbody as a function of wavelength. Furthermore, he was able to describe how this spectrum would change as the temperature changed. Planck's work on blackbody radiation is one of the areas of physics that led to the foundation of the wonderful science of Quantum Mechanics, but that is unfortunately beyond the scope of this article.

Planck and the others found was that as the temperature of a blackbody increases, the total amount of light emitted per second increases, and the wavelength of the spectrum's peak shifts to bluer colors .



For example, an iron bar becomes orange-red when heated to high temperatures and its color progressively shifts toward blue and white as it is heated further.

In 1893, German physicist Wilhelm Wien quantified the relationship between blackbody temperature and the wavelength of the spectral peak with the following equation:

$$\lambda_{max} \cdot T = 0.29 \text{ cm K}$$

where T is the temperature in Kelvin. Wien's law (also known as Wien's displacement law) states that the wavelength of maximum emission from a blackbody is inversely proportional to its temperature. This makes sense; shorter-wavelength (higher-frequency) light corresponds to higher-energy photons, which you would expect from a higher-temperature object.

For example, the sun has an average temperature of 5800 K, so its wavelength of maximum emission is given by:

$$\lambda_{max} = \frac{0.29 \text{ cm}}{5800} = 500 \text{ nm}$$

This wavelength falls in the green region of the visible light spectrum, but the sun's continuum radiates photons both longer and shorter than λ_{max} and the human eye perceives the sun's color as yellow/white.

In 1879, Austrian physicist Stephan Josef Stefan showed that the luminosity, L , of a black body is proportional to the 4th power of its temperature T .

$$L = A \cdot \alpha \cdot T^4$$

where A is the surface area, α is a constant of proportionality, and T is the temperature in Kelvin. That is, if we double the temperature (e.g. 1000 K to 2000 K) then the total energy radiated from a blackbody increases by a factor of 2^4 or 16.

Five years later, Austrian physicist Ludwig Boltzmann derived the same equation and is now known as the Stefan-Boltzmann law. If we assume a spherical star with radius R , then the luminosity of such a star is

$$L = 4\pi R^2 \cdot \alpha \cdot T^4$$

where R is the star radius in cm, and the α is the Stefan-Boltzmann constant, which has the value:

$$\alpha = 5.670 \cdot 10^{-8} \text{ W/m}^2\text{K}^4$$

Blackbody Radiation Heat Transfer

The heat emitted by a blackbody (per unit time) at an absolute temperature of T is given by the **Stefan-Boltzmann Law** of thermal radiation,

$$\dot{Q} = A\sigma T^4 = AE_{\text{blackbody}}$$

where $\dot{Q}$ has units of Watts, A is the total radiating area of the blackbody, and σ is the Stefan-Boltzmann constant.

A small blackbody at absolute temperature T enclosed by a much larger blackbody at absolute temperature T_e will transfer a **net heat flow** of,

$$\dot{Q} = A\sigma(T^4 - T_e^4)$$

The small blackbody still emits a total heat flow given by the Stefan-Boltzmann law. However, the small blackbody also receives and absorbs all the thermal energy emitted by the large enclosing

blackbody, which is a function of its temperature T_e . The difference in these two heat flows is the net heat flow lost by the small blackbody.

Gray Body Radiation Heat Transfer

Bodies that emit less thermal radiation than a blackbody have surface emissivities ϵ less than 1. If the surface emissivity is independent of wavelength, then the body is called a "gray" body, in that no particular wavelength (or color) is favored.

The net heat transfer from a small gray body at absolute temperature T with surface emissivity ϵ to a much larger enclosing gray (or black) body at absolute temperature T_e is given by,

$$\dot{Q} = \epsilon A \sigma (T^4 - T_e^4)$$

Radiation Exchange between small gray bodies:

Consider two gray bodies 1 and 2 having emissivities ϵ_1 , ϵ_2 or absorptivities α_1 , α_2 .

They are said to be small if their size is very small compared to the distance between them. The radiation emitted by 1 is partly absorbed by 2. The portion of radiation unabsorbed and thus reflected on the first incidence is considered to be lost in space, that is, nothing returns again to surface 1. The same can be said of surface 2 as well.

The energy emitted by body 1 = $A_1 \epsilon_1 \sigma T_1^4$

The energy incident on body 2 = $F_{12} A_2 \epsilon_1 \sigma T_1^4$

The energy absorbed by body 2 = $\alpha_2 F_{12} A_2 \epsilon_1 \sigma T_1^4$

The energy transfer from 1 to 2 is

$$Q_1 = \epsilon_1 \epsilon_2 A_1 F_{12} \sigma T_1^4$$

Similarly energy transfer from 2 to 1 is

$$Q_2 = \epsilon_1 \epsilon_2 A_2 F_{21} \sigma T_2^4$$

Net radiant heat exchange between the two bodies is

$$Q_{12} = \epsilon_1 \epsilon_2 A F \sigma (T_1^4 - T_2^4)$$

Where $AF = A_1 F_{12} = A_2 F_{21}$

Radiation exchange between large parallel gray planes

Consider two very large parallel gray surfaces A₁ and A₂, a small distance apart and exchanging radiation. All the radiation emitted by one plane must reach and be intercepted by the other plane.

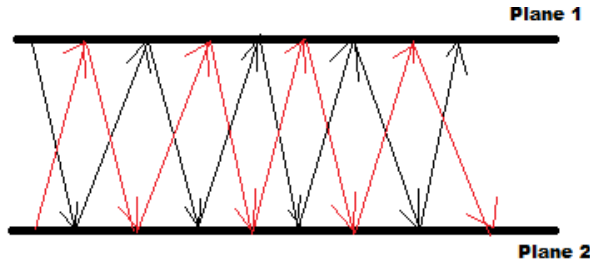


Fig.8. Heat exchange between two large parallel planes by radiation

Radiation emitted by A₁ = $\epsilon_1 \sigma T_1^4$

Radiation absorbed by A₂ = $\alpha_2 \epsilon_1 \sigma T_1^4$

(on first incidence)

Radiation reflected by A₂ = $\rho_2 \epsilon_1 \sigma T_1^4$

Radiation absorbed by A₁ = $\alpha_1 \rho_2 \epsilon_1 \sigma T_1^4$

Radiation reflected by A₁ = $\rho_1 \rho_2 \epsilon_1 \sigma T_1^4$

Radiation absorbed by A₂ = $\alpha_2 \rho_1 \rho_2 \epsilon_1 \sigma T_1^4 = \rho_1 \rho_2 \epsilon_1 \epsilon_2 \sigma T_1^4$

The same applies to surface 2.

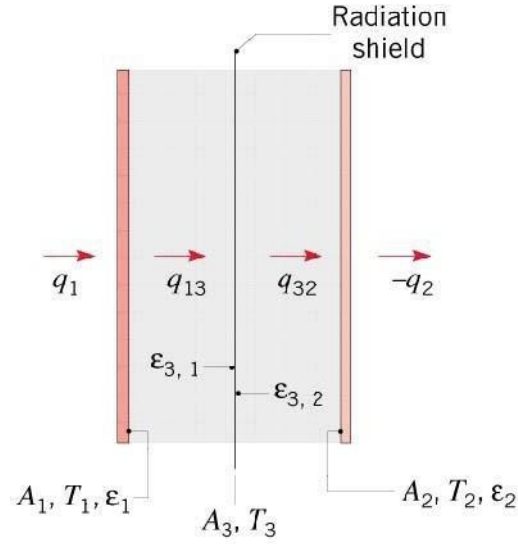
The net exchange of energy for an area A is

$$Q_{12} = \frac{\epsilon_1 \epsilon_2}{1 - \rho_1 \rho_2} A \sigma (T_1^4 - T_2^4) = \frac{1}{\left(\frac{1}{\epsilon_1}\right) + \left(\frac{1}{\epsilon_2}\right) - 1} A \sigma (T_1^4 - T_2^4)$$

Radiation shields

Radiation shields are often used to reduce the heat transfer by radiation between surfaces by effectively increasing the surface resistance without actually removing the heat from the overall system. A very effective insulation can be provided by using many layers of radiation reflecting films separated by a vacuum. Thin sheets of plastic coated with highly reflecting metallic films on both sides serve as very effective radiation shields. A familiar application of radiation shields is in the measurement of the temperature of a fluid by a thermometer or a thermocouple which is shielded to minimize the radiation effects.

Consider use of a single shield in a two-surface enclosure, such as that associated with large parallel plates:



$$Q_{13} = \frac{1}{\left(\frac{1}{\epsilon_1}\right) + \left(\frac{1}{\epsilon_3}\right) - 1} A \sigma (T_1^4 - T_3^4)$$

$$Q_{32} = \frac{1}{\left(\frac{1}{\epsilon_3}\right) + \left(\frac{1}{\epsilon_2}\right) - 1} A \sigma (T_3^4 - T_2^4)$$

$$Q_{12} = \frac{1}{\left(\frac{1}{\epsilon_1}\right) + \left(\frac{1}{\epsilon_2}\right) + 2\left(\frac{1}{\epsilon_3}\right) - 2} A \sigma (T_1^4 - T_2^4)$$

This equation can be generalized for a system of two parallel plates separated by n screens of emissivity as

$$Q_{12} = \frac{1}{\left(\frac{1}{\epsilon_1}\right) + \left(\frac{1}{\epsilon_2}\right) + 2\sum_{i=1}^n \left(\frac{1}{\epsilon_{si}}\right) - (n+1)} A \sigma (T_1^4 - T_2^4)$$

1. Calculate the following for an industrial furnace in the form of a black body and emitting radiation at 2500 °C

- (i) Monochromatic emissive power at 1.2 μm length
- (ii) Wavelength at which the emissive is maximum
- (iii) Maximum emissive powers
- (iv) Total emissive power.

Given Data:

Surface temperature, $T = 2500\text{ }^{\circ}\text{C} = 2773\text{ K}$

To Find :

- (i) Monochromatic emissive power, E_b at $\lambda = 1.2\text{ m}$ length
 $\lambda = 1.2 \times 10^{-6}\text{ m}$
- (ii) Wavelength at maximum, $\lambda_{\max}$
- (iii) Maximum emissive power, $(E_{b\lambda})_{\max}$
- (iv) Total emissive power, E_b

Solution:

$$(i) \quad E_{b\lambda} = \frac{C_1 \lambda^{-5}}{e^{\left[\frac{C_2}{\lambda T} \right]} - 1} \quad [\text{HMT data book. Pg. 82}]$$

$$\text{Let } C_1 = 0.374177107 \times 10^{-15} \text{ W/m}^2 \quad C_2 = 0.014387752 \text{ mk}$$

$$E_{b\lambda} = \frac{0.374177107 \times 10^{-15} \times (1.2 \times 10^{-6})^{-5}}{e^{\left[\frac{0.014387752}{1.2 \times 10^{-6} \times 2773} \right]} - 1} = \frac{1.5037 \times 10^{14}}{74.472}$$

$$E_{b\lambda} = 2.019 \times 10^{12} \text{ W/m}^2$$

$$(ii) \quad \lambda_{\max} T = 2898 \text{ } \mu\text{mk}$$

$$\lambda_{\max} \cdot 2773 = 2898 \times 10^{-6} \text{ mk}$$

$$\lambda_{\max} = 1.045 \times 10^{-6} \text{ m}$$

$$(iii) \quad (E_{b\lambda})_{\max} = 1.307 \times 10^{-5} T^5$$

$$= 1.307 \times 10^{-5} (2773)^5$$

$$(E_{b\lambda})_{\max} = 2.143 \times 10^{12} \text{ W/m}^2$$

$$(iv) \quad E_b = \sigma T^4$$

$$= 5.67 \times 10^{-8} \times (2773)^4$$

$$E_b = 3.352 \times 10^6 \text{ W/m}^2$$

2. Two parallel plates of size 1m x 1m spaced 0.5m apart are located in very large room, the walls are maintained at a temperature of 27°C. One plate is maintained at a temperature of 900°C and the other at 400°C. Their emissivities are 0.2 and 0.5 respectively. If the plates exchange heat between themselves and surroundings, find the heat transfer to each plate and to them. Consider only the plate surface facing each other. (16 marks) [May/June 2012]

Given Data:

Plate size = 1m x 1m

Plate distance = 0.5 m

First plate temperature, $T_1 = 900^\circ\text{C} = 1173\text{K}$

Second plate temperature, $T_2 = 400^\circ\text{C} = 673\text{K}$

Emissivity of first plate, $\epsilon_1 = 0.2$

Emissivity of second plate, $\epsilon_2 = 0.5$

To Find:

- Heat transfer to each plate
- Heat transfer to room.

Solution:

Heat transfer take place two plates and also room. So, this is three surface problem.

Area, $A_1 = 1 \times 1 = 1\text{m}^2$ $A_1 = A_2 = 1\text{m}^2$

The room size is infinity. so $A_3 = \infty$ [HMT data book pg.no.93]

$X = L/D = 1/0.5 = 2$

$Y = B/D = 1/0.5 = 2$

$F_{12} = 0.41525$

We know that, $F_{11} + F_{12} + F_{13} = 1$ But, $F_{11} = 0$ [Due to room]

We know that,

$F_{11} + F_{12} + F_{13} = 1$ But, $F_{11} = 0$

So, $0 + F_{12} + F_{13} = 1$

$F_{13} = 1 - F_{12}$

$= 1 - 0.41525$

$F_{13} = 0.5847$

Similarly,

$F_{21} + F_{22} + F_{23} = 1$

But, $F_{22} = 0$ [Due to room]

$F_{21} + 0 + F_{23} = 1$

$F_{23} = 1 - F_{21} = 1 - 0.41525$

$F_{23} = 0.5847$

Stefan – Boltzmann law, $E_b = \sigma T^4$

$E_{b1} = \sigma T_1^4$

$= 5.67 \times 10^{-8} (1173)^4$

$E_{b1} = 107.34 \times 10^3 \text{ W/m}^2$

$$E_{b2} = \sigma T_2^4 = 5.67 \times 10^{-8} (673)^4$$

$$E_{b2} = 11.63 \times 10^3 \text{ W/m}^2$$

$$E_{b3} = \sigma T_3^4 = 5.67 \times 10^{-8} (300)^4$$

$$E_{b3} = 459.27 \text{ W/m}^2$$

At Node J_1 ,

$$\frac{E_{b1} - J_1}{4} + \frac{J_2 - J_1}{\frac{1}{A_1 F_{12}}} + \frac{E_{b3} - J_1}{\frac{1}{A_1 F_{13}}} = 0$$

$$\frac{107.34 \times 10^3 - J_1}{4} + \frac{J_2 - J_1}{\frac{1}{1 \times 0.41525}} + \frac{459.27 - J_1}{\frac{1}{1 \times 0.5847}} = 0$$

$$26835 - 0.25 J_1 + 0.415 J_2 - 0.45 J_1 + 268.54 - 0.5847 J_1 = 0$$

$$-1.2497 J_1 + 0.415 J_2 = -27.10 \times 10^3 \longrightarrow \textcircled{1}$$

At Node J_2 ,

$$(J_1 - J_2) / (1/A_1 F_{12}) + (E_{b3} - J_2) / (1/A_2 F_{23}) + (E_{b2} - J_2) / 2 = 0$$

$$(J_1 - J_2) / (1/1 \times 0.41525) + (459.27 - J_2) / (1/1 \times 0.5847) + (11.63 \times 10^3 - J_2) / 2 = 0$$

$$J_1/2.408 - J_2/2.408 + 268.54 - 0.5847 J_2 + 5.815 \times 10^3 - 0.5 J_2 = 0$$

$$\Rightarrow 0.415 J_1 - 1.4997 J_2 = -6.08 \times 10^3 \longrightarrow \textcircled{2}$$

From solving Eqn. 1 & Eqn. 2

$$-1.2497 J_1 + 0.415 J_2 = -27.10 \times 10^3$$

$$0.415 J_1 - 1.4997 J_2 = -6.08 \times 10^3$$

$$\text{We get, } J_2 = 11.06 \times 10^3 \text{ W/m}^2$$

$$J_1 = 25.35 \times 10^3 \text{ W/m}^2$$

$$\text{Heat lost by plate 1, } Q_1 = \frac{E_{b1} - J_1}{\frac{1 - \epsilon_1}{A_1 \epsilon_1}} = \frac{107.34 \times 10^3 - 25.35 \times 10^3}{(1 - 0.2) / (1 \times 0.2)}$$

$$Q_1 = 20.49 \times 10^3 \text{ W}$$

$$\text{Heat lost by plate 2, } Q_2 = E_{b2} - J_2 = \frac{11.63 \times 10^3}{(1 - \epsilon_2)/A_2 \epsilon_2} - \frac{11.06 \times 10^3}{(1 - 0.5)/1 \times 0.5}$$

$$Q_2 = 570 \text{ W}$$

$$\text{Total heat lost, } Q = Q_1 + Q_2 = 20.49 \times 10^3 + 570$$

$$Q = 21.06 \times 10^3 \text{ W}$$

$$\text{Total heat received/absorbed by room, } Q = \frac{J_1 - J_3}{1/A_1 F_{13}} + \frac{J_2 - J_3}{1/A_2 F_{23}}$$

$$J_3 = E_{b3}$$

$$= \frac{25.35 \times 10^{-3} - 459.27}{1/1 \times 0.5847} + \frac{11.06 \times 10^3 - 459.27}{1/1 \times 0.5847}$$

$$Q = 20.752 \times 10^3 \text{ W}$$

3. Derive Wien's displacement law of radiation from plank's law. (May/June 2012)

Wien's displacement law states that the product of $\lambda_{\max}$ and T is constant

$$\text{(i.e.) } \lambda_{\max} \cdot T = \text{Constant}$$

From Planck's law,

$$(E_\lambda)_b = \frac{2\pi C^2 h \lambda^{-5}}{\left[e^{ch/\lambda KT} \right]^{-1}}$$

$(E_\lambda)_b$ becomes maximum (if T = Constant) when

$$\frac{d(E_\lambda)_b}{d\lambda} = 0$$

$$\frac{d(E_\lambda)_b}{d\lambda} = \frac{d}{d\lambda} \left[\frac{C_1 \lambda^{-5}}{\left[\frac{C_2}{\exp} \right] \lambda T} \right]^{-1} = 0 \quad (\text{or})$$

$$\frac{\left[\exp \left(\frac{C_2}{\lambda T} \right) \right]^{-1} \left(-5C_1 \lambda^{-6} \right) - \left\{ \left[\frac{C_1}{\exp} \right] \lambda^{-5} \left[\frac{C_2}{\lambda T} \right] \right\}}{e^{C_2/\lambda T} \lambda^{-1} \lambda^2} = 0$$

$$- \exp \left[-\frac{C_2}{\lambda T} \right] + 1 + \frac{1}{5} C_2 \left(\frac{1}{\lambda T} \right) \exp \left[-\frac{C_2}{\lambda T} \right] = 0$$

By Trial and error.

$$\lambda_{\max} T = \frac{C_2}{4.965} = \frac{1.439 \times 10^4}{4.965} \mu\text{mK}$$

$$\lambda_{\max} T = 2898 \mu\text{mK}$$

4. (a) Find The energy emitted by a black body at 700°C [Nov/Dec-2011]

GIVEN DATA:

Surface temperature, $T = 700^\circ\text{C} = 973\text{K}$.

To FIND:

Energy emitted by a black body

SOLUTION :

Energy emitted, $E_b = \sigma T^4$

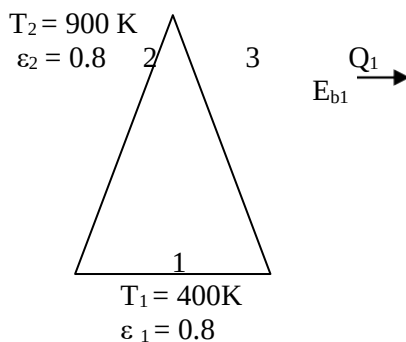
$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$$

HMT data book Pg.no .82

$$E_b = 5.67 \times 10^{-8} (973)^4 = 50.82 \times 10^3 \text{ W/m}^2$$

(b) A furnace is approximated as an equilateral triangular duct of sufficient length so that end effect can be neglected. The hot wall of the furnace is maintained at 900K and has the same emissivity. Find the net radiation heat blur leaving. The wall. Third wall of the furnace may be assured as a reradiating surface.

Two of the surfaces of a long equilateral triangular furnace are maintained at uniform temperature while the third wall is reradiating surface.



The radiation network in This Case is a simple series parallel Connection.

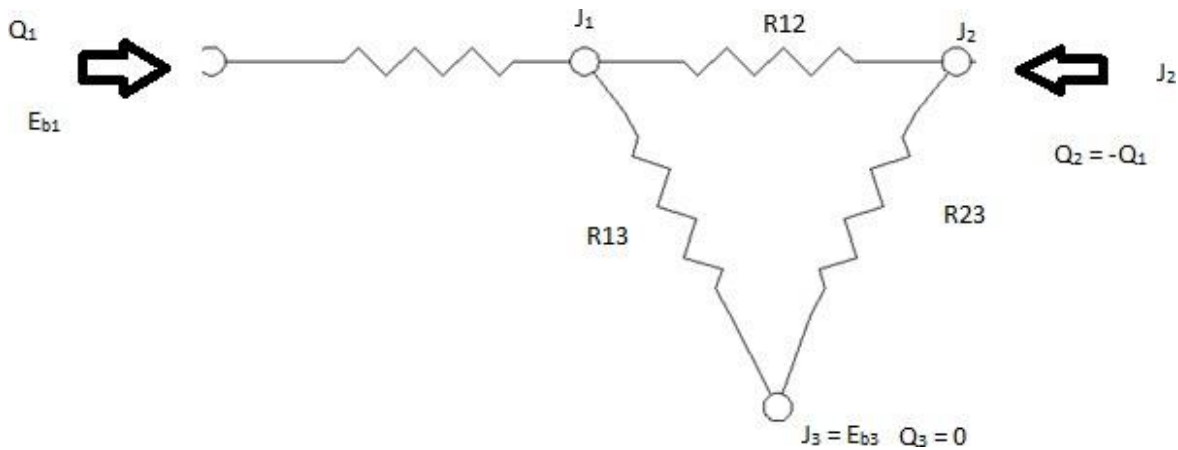
$$\frac{Q_1}{A} = \frac{E_{b1} - E_{b2}}{\left[\frac{1}{\epsilon_1} + \frac{1}{F_{12}} + \frac{1}{F_{13}} + \frac{1}{F_{23}} \right]}$$

$$\frac{Q_1}{A} = \frac{E_{b1} - E_{b2}}{\left[\frac{1 - \epsilon_1}{\epsilon_1} + \frac{1}{F_{12} + \frac{1}{\frac{1}{R_{13}} + \frac{1}{R_{23}}}} \right]} + \frac{1 - \epsilon_2}{\epsilon_2}$$

$$F_{12} = F_{13} = F_{23} = 0.5 \text{ [Symmetry]}$$

$$E_{b1} = \sigma T_1^4 = 5.67 \times 10^{-8} \times (400)^4 = 1451.52 \text{ W/m}^2$$

$$E_{b2} = \sigma T_2^4 = 5.67 \times 10^{-8} \times (900)^4 = 37200.87 \text{ W/m}^2$$



$$\frac{Q}{A} = \frac{37200.87 - 1451.52}{\left[\frac{1-0.8}{0.8} \right] + \left[\frac{0.5 + 1}{0.5 + 0.5} \right] + \left[\frac{1-0.8}{0.8} \right]}$$

$$\frac{Q}{A} = 17874.675 \text{ W/m}^2$$

1) a) Considering radiation in gases, obtain the exponential – decay Formula.

[Nov/Dec – 2011]

Consider a beam of monochromatic radiation with an intensity $I_{\lambda 0}$ entering a gas layer of Thickness L .

As the beam passes through The gas layer, its intensity gets reduce and the decrease is given by.

$$dI_{\lambda x} = -K_{\lambda} I_{\lambda x} \cdot dx$$

Where,

$I_{\lambda x}$ – Monochromatic intensity at a distance x

K_{λ} – Proportionality Constant

Integrating above equation between $x=0$ and $X=L$

$$\int_{x=0}^{x=L} \frac{dI_{\lambda x}}{I_{\lambda x}} = \int_{x=0}^{x=L} -K_{\lambda} \cdot dx$$

$$\ln \frac{I_{\lambda L}}{I_{\lambda 0}} = -K_{\lambda} \cdot L$$

$$I_{\lambda L} = I_{\lambda 0} e^{-K_{\lambda} \cdot L}$$

Where, $I_{\lambda L}$ – radiation intensity at $x = L$

5. b) Consider two concentric cylinders having diameters 10cm and 20cm and a length of 20cm. Designating The open ends of the cylinders as surface 3 and 4, estimate the shape factor, F_{3-4} (10 marks) [Nov/Dec – 2011]

From data book pg.no 104

$$r_3 = 5 \text{ cm}$$

$$r_4 = 10 \text{ cm}; L=20\text{cm}$$

$$L/r_4 = 20/10 = 2; r_3/r_4 = 5/10 = 0.5$$

$$F_{4-3} = 0.44$$

$$A_3 F_{3-4} = A_4 F_{4-3}$$

$$F_{3-4} = A_4/A_3 \cdot F_{4-3} = \frac{\pi \times 10^2 \times 0.44}{\pi \times 5^2} = 4 \times 0.44 = 1.76$$

$$\text{Shape factor, } F_{3-4} = 1.76$$

6. a) Distinguish between irradiation and radiosity ? [May/June 2013]

Radiation intensity is nothing but radiation heat transfer. The heat radiated per unit time per unit area is The radiation intensity ($Q/A = q$)

Irradiation: G = Total radiation incident upon a surface per unit time and per unit area.

$$Q/A = J - G \text{ (or) } Q = \frac{E_b - J}{(1-\epsilon)/\epsilon A}$$

b) A double walled cylindrical vessel used for storing liquid oxygen at -18.c The inner wall is coated with a paint with $\epsilon = 0.02$. The temperature of the inner surface of the outer wall is 2.c. It is also coated with same painting. The inner wall surface area is 80% of outer the inner wall surface area is 0.15 m². Determine heat radiated. [Apr/May 2012]

SOLUTION :

$$A_1 = 0.15 \text{ m}^2$$

$$0.15 = 0.8 A_2$$

$$A_2 = 0.15/0.8 = 0.1875 \text{ m}^2$$

$$R_1 = \frac{1-\epsilon_1}{A_1 \epsilon_2} = \frac{1-0.02}{0.15 \times 0.02} = 326.66 \text{ 1/m}^2$$

$$R_2 = \frac{1-\epsilon_2}{A_2 \epsilon_2} = \frac{1-0.02}{0.1875 \times 0.02} = 261.33 \text{ 1/m}^2$$

$$F_{12} = 1 \text{ [Since "L" is not given, } L = \infty, \text{ Refer page 84 of HMT data book]}$$

$$R = \frac{1}{A_1 F_{12}} = \frac{1}{0.15 \times 1} = 6.666 \text{ /m}^2$$

$$E_{b1} - E_{b2} = \sigma (T_1^4 - T_2^4) = 5.67 \times 10^{-8} (255^4 - 293^4) = -178.14 \text{ W/m}^2$$

$$Q_{12} = \frac{E_{b1} - E_{b2}}{R_1 + R + R_2} = \frac{-178.14}{326.66 + 6.66 + 261.33}$$

$$Q_{12} = -0.29956 \text{ W}$$

“-ve sign indicates that heat is radiated in opposite direction.

7. Two very large parallel planes are respectively. 0.8 and 0.3. To minimize the radiation exchange between the planes, a polished aluminum radiation shield is placed between them. If the emissivity of the shield is 0.04 on both sides, find the percentage reduction in heat transfer rate. (10 marks)[Apr/May – 2011]

GIVEN DATA:

Emissivity, $\epsilon_1 = 0.8$, $\epsilon_2 = 0.3$, $\epsilon_3 = 0.04$

To find :

% Reduction in heat transfer rate.

Solution:

Heat transfer without radiation shield.

$$Q_{12} = \bar{\epsilon} \sigma A (T_1^4 - T_2^4)$$

$$\text{Let } \bar{\epsilon} = \frac{1}{1/\epsilon_1 + 1/\epsilon_2 - 1} = \frac{1}{1/0.8 + 1/0.3 - 1}$$

$$\bar{\epsilon} = 0.279$$

$$Q_{12} = 0.279 \sigma A (T_1^4 - T_2^4) \longrightarrow \text{---} 1$$

Heat Transfer with radiation shield

$$Q_{\text{radiation shield}} = \frac{A \sigma (T_1^4 - T_2^4)}{1/\epsilon_1 + 1/\epsilon_2 + 2n/\epsilon_3 - (n+1)}$$

Let n = No. of radiation shield. $Q_{\text{radiation shield}}$

$$\begin{aligned} Q_{\text{radiation shield}} &= \frac{A \sigma (T_1^4 - T_2^4)}{1/0.8 + 1/0.3 + 2 \times 1/0.04 - (1+1)} \\ &= \frac{A \sigma [T_1^4 - T_2^4]}{52.58} \end{aligned}$$

$$Q_{\text{radiation shield}} = 0.019 A \sigma (T_1^4 - T_2^4)$$

$$\begin{aligned} \% \text{ Reduction in heat transfer rate} &= \frac{Q_{\text{without shield}} - Q_{\text{with shield}}}{Q_{\text{without shield}}} \\ &= \frac{0.279 \sigma A (T_1^4 - T_2^4) - 0.019 A \sigma (T_1^4 - T_2^4)}{0.279 \sigma A (T_1^4 - T_2^4)} \\ &= \frac{0.279 - 0.019}{0.279} = 93.18\% \end{aligned}$$

8. Define the Following [Nov/Dec – 2010]

(i) Black body (ii) Grey body (iii) Opaque body (iv) white body

(i) Black body :

This black body absorbs all the radiation falling on its surface.

For this body $\alpha = 1$, So $\tau = 0$, $\rho = 0$

(ii) Grey body :

For this grey body, absorptivity does not vary with temperature and wave length of the incident radiation. So, for a grey body $\alpha = (\alpha) \lambda = \text{Constant}$.

(iii) Opaque body :

This opaque body does not transmit any radiation falling on its surface. Here $\tau = 0$, So, $\alpha + \rho = 1$

(iv) White body:

This white body reflects all the radiation falling on its surfaces

9. The radiation shape factor of the circular surface of a thin hollow cylinder of 10cm diameter and 10cm length is 0.1716. What respect to itself ? [Nov/Dec – 2010]

Given data :

$$r_1 = r_2 = 10/2 = 5\text{cm}, L = 10\text{cm}, F_{1-2} = 0.1716$$

$$\text{Here } F_{1-2} = F_{2-1} \text{ as } A_1 = A_2$$

The shape factor relation between all the three surfaces is given by,

$$F_{1-1} + F_{1-2} + F_{1-3} = 1 \quad \longrightarrow \quad (1)$$

$$F_{3-3} + F_{3-2} + F_{3-1} = 1 \quad \longrightarrow \quad (2)$$

$$\text{But } F_{3-1} = F_{3-2} \text{ and } F_{1-1} = F_{2-2} = 0 \quad \longrightarrow \quad (3)$$

Substituting (3) in (1)

$$F_{3-3} + F_{3-1} + F_{3-1} = 1$$

$$F_{3-3} = 1 - 2 F_{3-1} \quad \longrightarrow \quad (4)$$

$$\text{Also } A_1 F_{1-3} = A_3 F_{3-1}$$

$$F_{3-1} = F_{3-1} \times A_1/A_3 = F_{1-3} \times \frac{\pi R^2}{2\pi r L}$$

$$F_{3-1} = F_{1-3} \times r/2L$$

From equation (1):

$$F_{1-3} = 1 - F_{1-2} (F_{1-1} = 0) = 1 - 0.1716 = 0.8284 \quad \longrightarrow \quad (5)$$

$$F_{3-1} = 0.8284 \times 5/(2 \times 10) = 0.2071$$

Now substituting F_{3-1} in eqn. (4)

$$F_{3-3} = 1 - 2 (0.2071) = 0.5858$$

10. (a) Discuss briefly the variation of black body emissive power with wave length for different temperatures [Apr/May – 2008]

A plot of $(E_\lambda)_b$ as a function of temperature and wavelength is given in

The plot shows the following characteristics of block body radiations.

- 1) The energy emitted at all wavelength increases with rise in temperature.
- 2) The peak spectral emissive power shift towards a smaller, wavelength at higher temperature.
- 3) The area under the monochromatic emissive power versus wavelength at any temperature. Given the rate of radiation energy emitted within the wavelength interval $d\lambda$.

$$dE_b = (E_\lambda)_b \cdot d\lambda.$$

b) The spectral emissivity function of an opaque surface at 800K is approximated as .

$$\varepsilon_\lambda = \begin{cases} \varepsilon_1 = 0.3 & 0 \leq \lambda < 3\mu\text{m} \\ \varepsilon_2 = 0.8 & 3\mu\text{m} \leq \lambda < 7\mu\text{m} \\ \varepsilon_3 = 0.1 & 7\mu\text{m} \leq \lambda < \infty \end{cases}$$

Calculate the average emissivity of the surface and its emissive power.

}

Average emissivity of the surface,

$$\varepsilon = \varepsilon_1 \varepsilon_2 + \varepsilon_2 \varepsilon_3 + \varepsilon_3 \varepsilon_1$$

$$= (0.3 \times 0.8) + (0.8 \times 0.1) + (0.1 \times 0.3) = 0.35$$

Emissive power, $E_b = \sigma \varepsilon T^4$ (or) $\varepsilon \sigma T^4$

$$= 0.35 \times 5.67 \times 10^{-8} \times 800^4$$

$$E_b = 8128.5 \text{ W/m}^2$$

11) Explain briefly the following

[April/May – 2008]

(i) Specular and diffuse reflection

(ii) Reflectivity & transmissivity

(iii) Reciprocity rule and summation rule

(i) Specular and diffuse reflection :

$$Q_{12} = \frac{1}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1} + \frac{1}{\frac{1}{\epsilon_3} + \frac{1}{\epsilon_2} - 1}$$

If $\epsilon_1 = \epsilon_2 = \epsilon_3$, Then right side of equation (E) reduces to $\frac{1}{2}$ (or)

$$Q_{13} = Q_{32} = \frac{1}{2} Q_{12}$$

Thus by inserting one shield between two parallel surface, the direct radiation heat transfer between them is halved.

Sometime more than one shield is used, and in general case where there are n shields, all the surface resistance would be the same, since the emissivities are equal. There will be two of these surface resistance for each shield and one for each heat transfer surface, there will also be $(n+1)$ shape resistance but the shape factors are unity for all infinite parallel planes.

$$\text{Total Resistance (R}_n\text{)} = \frac{(2n+2) \left(\frac{1-\epsilon}{\epsilon} \right) + (n+1) (1)}{A} = \frac{(n+1) \left(\frac{2}{\epsilon} - 1 \right)}{A}$$

The radiant heat transfer rate between two infinitely large parallel plates separated by n shields is therefore.

$$Q_n = \frac{1 \cdot A \cdot \sigma (T_1^4 - T_2^4)}{(n+1) \left(\frac{2}{\epsilon} - 1 \right)}$$

12.a) Emissivities of two parallel plates at 800.C and 300.C are 0.3 and 0.5 respectively. Find the net energy transfer rate per square meter. [Nov/Dec – 2007]

SOLUTION :

The heat exchange per unit area between two large parallel plates is given by.

$$\begin{aligned} \frac{Q_{12}}{A} = q_{12} &= \frac{\sigma(T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{5.67 \times 10^{-8} [1073^4 - 573^4]}{\frac{1}{0.3} + \frac{1}{0.5} - 1} \\ &= \frac{5.67 \times 12352}{4.33} = 16.2 \text{ KW/m}^2 \end{aligned}$$

b) Show from energy balance consideration that the radiation heat transfer from a plane Composite surface area A_4 and made up of plane surface area A_2 and A_3 to plane surface area A_1 is given by,

$$A_4 F_{41} = A_3 F_{31} + A_2 F_{21}, \text{ and } F_{14} + F_{12} + F_{13}$$

PROOF :

If two surface A_1 and A_4

Are parallel and large, radiation

Occurs across the gap between them

so that A_1 and A_4 and all

radiation emitted by one falls on

the other, then

$$F_{1-4} = F_{4-1} = 1$$

If one of the surface say A_4 is divided into subareas A_2 and A_3 then $A_1 F_{1-2} = A_2 F_{12} + A_3 F_{13}$

Thus if the radiant surface is subdivided, the shape factor from that surface with respect to the receiving surface is not equal to sum of individual shape factors.

$$A_4 F_{4-1} = A_3 F_{31} + A_2 F_{21} \text{ (or) } F_{41} = F_{31} + F_{21}$$

Hence the shape factor from a radiating surface to a subdivided receiving surface is simply the sum of individual shape factors.

c) Using the definition of radiosity and irradiation prove that the radiation heat exchange between two grey bodies is given by the radiation

$$Q_{\text{net}} = \frac{\sigma(T_{14} - T_{24})}{\frac{1-E_1}{A_1 E_1} + \frac{1}{A_1 F_{1-2}} + \frac{1-E_2}{A_2 E_2}}$$

The radiosity comprises the original emission from the surface plus the reflected portion of any radiation incident upon it.

$$J = E + \rho G \text{ (or) } J = E_b + \rho G$$

Where E_b = Emissive power of a perfect black body at the same temperature.

Also $\alpha + \rho + \tau = 1$

$$\alpha + \rho = 1 \text{ (} \tau = 0, \text{ the surface is black and opaque)}$$

$$\text{(or) } \rho = 1 - \alpha$$

$$J = \epsilon E_b + (1 - \alpha) G$$

But $\alpha = \epsilon$ (by Kirchhoff's law)

$$\text{(or) } J = \epsilon E_b + (1 - \epsilon) G$$

$$\text{(or) } G = \frac{J - \epsilon E_b}{1 - \epsilon}$$

The net energy leaving a surface is the difference between its radiosity and irradiation, Thus.

$$\begin{aligned} \frac{Q_{\text{net}}}{A} &= J - G = J - \frac{J - \epsilon E_b}{1 - \epsilon} = \frac{J(1 - \epsilon) - (J - \epsilon E_b)}{1 - \epsilon} \\ &= \frac{J - J\epsilon - J + \epsilon E_b}{1 - \epsilon} = \frac{\epsilon[E_b - J]}{1 - \epsilon} \\ Q_{\text{net}} &= \frac{A\epsilon(E_b - J)}{1 - \epsilon} = \frac{E_b - J}{1 - \epsilon / A\epsilon} \end{aligned}$$

The quantity $1-\epsilon/A\epsilon$ is called **surface resistance**.

Now consider the exchange of radiant energy between two surface 1 and 2

The net interchange of heat between the surface (Q_{12}) is given by $Q_{12} = J_1 A_1 F_{1-2} - J_2 A_2 F_{2-1}$

$$\text{But } A_1 F_{1-2} = A_2 F_{2-1}$$

$$Q_{12} = A_1 F_{1-2} (J_1 - J_2)$$

$$(\text{or}) Q_{1-2} = J_1 - J_2 / 1/A_1 F_{12}$$

1 is called space resistance.

$$\overline{A_1 F_{1-2}}$$

The net heat exchange between two gray surface is given by

$$\frac{1-\epsilon_1}{A_1 \epsilon_1} \quad \frac{1}{A_1 F_{1-2}} \quad \frac{1-\epsilon_2}{A_2 \epsilon_2}$$

$$(Q_{12})_{\text{net}} = \frac{E_{b1} - E_{b2}}{\frac{1-\epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{1-2}} + \frac{1-\epsilon_2}{A_2 \epsilon_2}} = \frac{\sigma(T_{14} - T_{24})}{\frac{1-\epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{1-2}} + \frac{1-\epsilon_2}{A_2 \epsilon_2}}$$

13. A black body at 3000 K emits radiation. Calculate the following:

- Monochromatic emissive power at 7 μm wave length.**
- Wave length at which emission is maximum.**
- Maximum emissive power.**
- Total emissive power,**
- Calculate the total emissive of the furnace if it is assumed as a real surface having emissivity equal to 0.85.**

Given: Surface temperature $T = 3000\text{K}$

Solution: 1. Monochromatic Emissive Power:

From Planck's distribution law, we know

$$E_{b\lambda} = \frac{C_1 \lambda^{-5}}{e^{\left(\frac{C_2}{\lambda T}\right)} - 1}$$

[From HMT data book, Page No.71]

Where

$$C_1 = 0.374 \times 10^{-15} \text{ W m}^2$$

$$C_2 = 14.4 \times 10^{-3} \text{ mK}$$

$$\lambda = 1 \times 10^{-6} \text{ m} \quad [\text{Given}]$$

$$\Rightarrow E_{b\lambda} = \frac{0.374 \times 10^{-15} [1 \times 10^{-6}]^{-5}}{\left[\frac{144 \times 10^{-3}}{1 \times 10^{-6} \times 3000} \right] - 1}$$

$$E_{b\lambda} = 3.10 \times 10^{12} \text{ W/m}^2$$

2. Maximum wave length (λ_{max})

From Wien's law, we know

$$\lambda_{\text{max}} T = 2.9 \times 10^{-3} \text{ mK}$$

$$\Rightarrow \lambda_{\text{max}} = \frac{2.9 \times 10^{-3}}{3000}$$

$$\lambda_{\text{max}} = 0.966 \times 10^{-6} \text{ m}$$

3. Maximum emissive power ($E_{b\lambda}$) max:

Maximum emissive power

$$(E_{b\lambda})_{\text{max}} = 1.307 \times 10^{-5} T^5$$

$$= 1.307 \times 10^{-5} \times (3000)^5$$

$$(E_{b\lambda})_{\text{max}} = 3.17 \times 10^{12} \text{ W/m}^2$$

4. Total emissive power (E_b):

From Stefan – Boltzmann law, we know that

$$E_b = \sigma T^4$$

[From HMT data book Page No.71]

Where σ = Stefan – Boltzmann constant

$$= 5.67 \times 10^{-8} \text{ W/m}^2 \text{K}^4$$

$$\Rightarrow E_b = (5.67 \times 10^{-8}) (3000)^4$$

$$E_b = 4.59 \times 10^6 \text{ W/m}^2$$

5. Total emissive power of a real surface:

$$(E_b)_{\text{real}} = \epsilon \sigma T^4$$

Where ϵ = Emissivity = 0.85

$$(E_b)_{\text{real}} = 0.85 \times 5.67 \times 10^{-8} \times (3000)^4$$

$$(E_b)_{\text{real}} = 3.90 \times 10^6 \text{ W / m}^2$$

14. Assuming sun to be black body emitting radiation at 6000 K at a mean distance of 12×10^{10} m from the earth. The diameter of the sun is 1.5×10^9 m and that of the earth is 13.2×10^6 m.

Calculation the following.

1. Total energy emitted by the sun.
2. The emission received per m^2 just outside the earth's atmosphere.
3. The total energy received by the earth if no radiation is blocked by the earth's atmosphere.
4. The energy received by a 2×2 m solar collector whose normal is inclined at 45° to the sun. The energy loss through the atmosphere is 50% and the diffuse radiation is 20% of direct radiation.

Given: Surface temperature $T = 6000$ K

Distance between earth and sun $R = 12 \times 10^{10}$ m

Diameter on the sun $D_1 = 1.5 \times 10^9$ m

Diameter of the earth $D_2 = 13.2 \times 10^6$ m

Solution:1. Energy emitted by sun $E_b = \sigma T^4$

$$\Rightarrow E_b = 5.67 \times 10^{-8} \times (6000)^4$$

[$\because \sigma = \text{Stefan - Boltzmann constant}$
 $= 5.67 \times 10^{-8} \text{ W /m}^2 \text{ K}^4$]

$$E_b = 73.4 \times 10^6 \text{ W/m}^2$$

Area of sun $A_1 = 4\pi R_1^2$

$$= 4\pi \times \left(\frac{1.5 \times 10^9}{2} \right)^2$$

$A_1 = 7 \times 10^{18} \text{ m}^2$

$\Rightarrow$ Energy emitted by the sun

$$E_b = 73.4 \times 10^6 \times 7 \times 10^{18}$$

$E_b = 5.14 \times 10^{26} \text{ W}$

2. The emission received per m^2 just outside the earth's atmosphere:

The distance between earth and sun $R = 12 \times 10^{10}$ m

$$\begin{aligned}\text{Area, } A &= 4\pi R^2 \\ &= 4 \times \pi \times (12 \times 10^{10})^2 \\ \boxed{A} &= \boxed{1.80 \times 10^{23} \text{ m}^2}\end{aligned}$$

$\Rightarrow$ The radiation received outside the earth atmosphere per m^2

$$\begin{aligned}&= \frac{E_b}{A} \\ &= \frac{5.14 \times 10^{26}}{1.80 \times 10^{23}} \\ &= 2855.5 \text{ W/m}^2\end{aligned}$$

3. Energy received by the earth:

$$\begin{aligned}\text{Earth area} &= \frac{\pi}{4} (D)^2 \\ &= \frac{\pi}{4} \times [13.2 \times 10^6]^2 \\ \boxed{\text{Earth area}} &= \boxed{1.36 \times 10^{14} \text{ m}^2}\end{aligned}$$

Energy received by the earth

$$\begin{aligned}&= 2855.5 \times 1.36 \times 10^{14} \\ &= 3.88 \times 10^{17} \text{ W}\end{aligned}$$

4. The energy received by a $2 \times 2 \text{ m}$ solar collector;

Energy loss through the atmosphere is 50%. So energy reaching the earth.

$$\begin{aligned}&= 100 - 50 = 50\% \\ &= 0.50\end{aligned}$$

Energy received by the earth

$$\begin{aligned}&= 0.50 \times 2855.5 \\ &= 1427.7 \text{ W/m}^2 \quad \dots\dots(1)\end{aligned}$$

Diffuse radiation is 20%

$$\begin{aligned}\Rightarrow 0.20 \times 1427.7 &= 285.5 \text{ W/m}^2 \\ \boxed{\text{Diffuse radiation}} &= \boxed{285.5 \text{ W/m}^2} \quad \dots\dots\dots(2)\end{aligned}$$

Total radiation reaching the collection

$$\begin{aligned}&= 1427.7 + 285.5 \\ &= 1713.2 \text{ W/m}^2\end{aligned}$$

$$\begin{aligned}\text{Plate area} &= A \times \cos \theta \\ &= 2 \times 2 \times \cos 45^\circ \\ &= 2.82 \text{ m}^2\end{aligned}$$

Energy received by the collector

$$= 2.82 \times 1713.2$$

$$= 4831.2 \text{ W}$$

15. Two black square plates of size 2 by 2 m are placed parallel to each other at a distance of 0.5 m. One plate is maintained at a temperature of 1000°C and the other at 500°C. Find the heat exchange between the plates.

Given: Area $A = 2 \times 2 = 4 \text{ m}^2$

$$T_1 = 1000^\circ\text{C} + 273$$

$$= 1273 \text{ K}$$

$$T_2 = 500^\circ\text{C} + 273$$

$$= 773 \text{ K}$$

$$\text{Distance} = 0.5 \text{ m}$$

To find : Heat transfer (Q)

Solution : We know Heat transfer general equation is

$$Q_{12} = \frac{\sigma [T_1^4 - T_2^4]}{\frac{1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1}{A_2 \epsilon_2}}$$

where [From equation No.(6)]

For black body $\epsilon_1 = \epsilon_2 = 1$

$$\Rightarrow Q_{12} = \sigma [T_1^4 - T_2^4] \times A F_{12}$$

$$= 5.67 \times 10^{-8} [(1273)^4 - (773)^4] \times 4 \times F_{12}$$

$$\boxed{Q_{12} = 5.14 \times 10^5 F_{12}} \quad \text{.....(1)}$$

Where F_{12} – Shape factor for square plates

In order to find shape factor F_{12} , refer HMT data book, Page No.76.

$$X \text{ axis} = \frac{\text{Smaller side}}{\text{Distance between planes}}$$

$$= \frac{2}{0.5}$$

$$\boxed{X \text{ axis} = 4}$$

Curve $\rightarrow 2$ [Since given is square plates]

X axis value is 4, curve is 2. So corresponding Y axis value is 0.62.

i.e., $\boxed{F_{12} = 0.62}$

$$(1) \Rightarrow Q_{12} = 5.14 \times 10^5 \times 0.62$$

$$\boxed{Q_{12} = 3.18 \times 10^5 \text{ W}}$$

16. Two parallel plates of size $3 \text{ m} \times 2 \text{ m}$ are placed parallel to each other at a distance of 1 m . One plate is maintained at a temperature of 550°C and the other at 250°C and the emissivities are 0.35 and 0.55 respectively. The plates are located in a large room whose walls are at 35°C . If the plates located exchange heat with each other and with the room, calculate.

1. Heat lost by the plates.

2. Heat received by the room.

Given: Size of the plates $= 3 \text{ m} \times 2 \text{ m}$

Distance between plates $= 1 \text{ m}$

First plate temperature $T_1 = 550^\circ\text{C} + 273 = 823 \text{ K}$

Second plate temperature $T_2 = 250^\circ\text{C} + 273 = 523 \text{ K}$

Emissivity of first plate $\varepsilon_1 = 0.35$

Emissivity of second plate $\varepsilon_2 = 0.55$

Room temperature $T_3 = 35^\circ\text{C} + 273 = 308 \text{ K}$

To find: 1. Heat lost by the plates

2. Heat received by the room.

Solution: In this problem, heat exchange takes place between two plates and the room. So this is three surface problems and the corresponding radiation network is given below. Area $A_1 = 3 \times 2 =$

$$6 \text{ m}^2 \quad \boxed{A_1 = A_2 = 6 \text{ m}^2}$$

Since the room is large $\boxed{A_3 = \infty}$

From electrical network diagram,

$$\frac{1 - \varepsilon_1}{\varepsilon_1 A_1} = \frac{1 - 0.35}{0.35 \times 6} = 0.309$$

$$\frac{1 - \varepsilon_2}{\varepsilon_2 A_2} = \frac{1 - 0.55}{0.55 \times 6} = 0.136$$

$$\frac{1 - \varepsilon_3}{\varepsilon_3 A_3} = 0 \quad [\because A_3 = \infty]$$

$$\text{Apply } \frac{1 - \varepsilon_3}{\varepsilon_3 A_3} = 0, \frac{1 - \varepsilon_1}{\varepsilon_1 A_1} = 0.309, \frac{1 - \varepsilon_2}{\varepsilon_2 A_2} = 0.136 \quad \text{values in electrical network diagram.}$$

To find shape factor F_{12} refer HMT data book, Page No.78.

$$X = \frac{b}{c} = \frac{3}{1} = 3$$

$$Y = \frac{a}{c} = \frac{2}{1} = 2$$

X value is 3, Y value is 2, corresponding shape factor

[From table]

$$F_{12} = 0.47$$

$$\boxed{F_{12} = 0.47}$$

We know that,

$$F_{11} + F_{12} + F_{13} = 1 \quad \text{But,} \quad F_{11} = 0$$

$$\Rightarrow F_{13} = 1 - F_{12}$$

$$\Rightarrow F_{13} = 1 - 0.47$$

$$\boxed{F_{13} = 0.53}$$

$$\text{Similarly, } F_{21} + F_{22} + F_{23} = 1$$

We know

$$F_{22} = 0$$

$$\Rightarrow F_{23} = 1 - F_{21}$$

$$\Rightarrow F_{23} = 1 - F_{12}$$

$$F_{13} = 1 - 0.47$$

$$\boxed{F_{23} = 0.53}$$

From electrical network diagram,

$$\frac{1}{A_1 F_{13}} = \frac{1}{6 \times 0.53} = 0.314 \quad \dots(1)$$

$$\frac{1}{A_2 F_{23}} = \frac{1}{6 \times 0.53} = 0.314 \quad \dots(2)$$

$$\frac{1}{A_1 F_{12}} = \frac{1}{6 \times 0.47} = 0.354 \quad \dots(3)$$

From Stefan – Boltzmann law, we know

$$E_b = \sigma T^4$$

$$E_{b1} = \sigma T_1^4$$

$$= 5.67 \times 10^{-8} [823]^4$$

$$\boxed{E_{b1} = 26.01 \times 10^3 \text{ W / m}^2} \quad \dots(4)$$

$$E_{b2} = \sigma T_2^4$$

$$= 5.67 \times 10^{-8} [823]^4$$

$$\boxed{E_{b2} = 4.24 \times 10^3 \text{ W / m}^2} \quad \dots(5)$$

$$E_{b3} = \sigma T_3^4$$

$$= 5.67 \times 10^{-8} [308]^4$$

$$\boxed{E_{b3} = J_3 = 510.25 \text{ W / m}^2} \quad \dots(6)$$

[From diagram]

The radiosities, J_1 and J_2 can be calculated by using Kirchoff's law.

$\Rightarrow$ The sum of current entering the node J_1 is zero.

At Node J_1 :

$$\frac{E_{b1} - J_1}{0.309} + \frac{J_2 - J_1}{\frac{1}{A_1 F_{12}}} + \frac{E_{b3} - J_1}{\frac{1}{A_1 F_{13}}} = 0$$

[From diagram]

$$\begin{aligned} \Rightarrow \frac{26.01 \times 10^3 - J_1}{0.354} + \frac{J_2 - J_1}{0.314} + \frac{510.25 - J_1}{0.309} &= 0 \\ \Rightarrow 84.17 \times 10^3 - \frac{J_1}{0.309} + \frac{J_2}{0.354} + \frac{J_1}{0.354} + 1625 - \frac{J_1}{0.354} &= 0 \\ \Rightarrow -9.24J_1 + 2.82J_2 &= -85.79 \times 10^3 \quad \text{.....(7)} \end{aligned}$$

At node j_2

$$\begin{aligned} \frac{J_1 - J_2}{\frac{1}{A_1 F_{12}}} + \frac{E_{b3} - J_2}{\frac{1}{A_2 F_{23}}} + \frac{E_{b2} - J_2}{0.136} &= 0 \\ \frac{J_1 - J_2}{0.314} + \frac{510.25 - J_2}{0.136} + \frac{4.24 \times 10^3 - J_2}{0.354} &= 0 \\ \Rightarrow \frac{J_1}{0.354} - \frac{J_2}{0.354} - \frac{J_2}{0.314} + \frac{510.25}{0.314} + \frac{4.24 \times 10^3}{0.136} - \frac{J_2}{0.136} &= 0 \\ \Rightarrow 2.82J_1 - 13.3J_2 &= -32.8 \times 10^3 \quad \text{....(8)} \end{aligned}$$

Solving equation (7) and (8),

$$\begin{aligned} \Rightarrow -9.24J_1 + 2.82J_2 &= -85.79 \times 10^3 \quad \text{.....(7)} \\ \Rightarrow 2.82J_1 - 13.3J_2 &= -32.8 \times 10^3 \quad \text{.....(8)} \end{aligned}$$

$$\begin{aligned} J_2 &= 4.73 \times 10^3 \text{ W / m}^2 \\ J_1 &= 10.73 \times 10^3 \text{ W / m}^2 \end{aligned}$$

Heat lost by plate (1) is given by

$$Q_1 = \frac{E_{b1} - J_1}{\left(\frac{1 - \epsilon_1}{\epsilon_1 A_1} \right)}$$

$$Q_1 = \frac{26.01 \times 10^3 - 10.73 \times 10^3}{\frac{1 - 0.35}{0.35 \times 6}}$$

$$Q_1 = 49.36 \times 10^3 \text{ W}$$

Heat lost by plate 2 is given by

$$Q_2 = \frac{E_{b2} - J_2}{\left(\frac{1 - \epsilon_2}{\epsilon_2 A_2} \right)}$$

$$Q_2 = \frac{4.24 \times 10^3 - 4.73 \times 10^3}{\frac{1-0.55}{6 \times 0.55}}$$

$$\boxed{Q_2 = -3.59 \times 10^3 \text{ W}}$$

Total heat lost by the plates

$$\begin{aligned} Q &= Q_1 + Q_2 \\ &= 49.36 \times 10^3 - 3.59 \times 10^3 \end{aligned}$$

$$\boxed{Q = 45.76 \times 10^3 \text{ W}} \quad \text{.....(9)}$$

Heat received by the room

$$\begin{aligned} Q &= \frac{J_1 - J_3}{\frac{1}{A_1 F_{13}}} + \frac{J_2 - J_3}{\frac{1}{A_1 F_{12}}} \\ &= \frac{10.73 \times 10^3 - 510.25}{0.314} = \frac{4.24 \times 10^3 - 510.25}{0.314} \\ &\quad [\because E_{b1} = J_1 = 512.9] \end{aligned}$$

$$\boxed{Q = 45.9 \times 10^3 \text{ W}} \quad \text{.....(10)}$$

From equation (9), (10), we came to know heat lost by the plates is equal to heat received by the room.

17. A gas mixture contains 20% CO₂ and 10% H₂O by volume. The total pressure is 2 atm. The temperature of the gas is 927°C. The mean beam length is 0.3 m. Calculate the emissivity of the mixture.

Given : Partial pressure of CO₂, $P_{\text{CO}_2} = 20\% = 0.20 \text{ atm}$

Partial pressure of H₂O,

$$P_{\text{H}_2\text{O}} = 10\% = 0.10 \text{ atm.}$$

Total pressure P = 2 atm

$$\begin{aligned} \text{Temperature } T &= 927^\circ\text{C} + 273 \\ &= 1200 \text{ K} \end{aligned}$$

Mean beam length $L_m = 0.3 \text{ m}$

To find: Emissivity of mixture (ϵ_{mix}).

Solution : To find emissivity of CO₂

$$P_{\text{CO}_2} \times L_m = 0.2 \times 0.3$$

$$\boxed{P_{\text{CO}_2} \times L_m = 0.06 \text{ m - atm}}$$

From HMT data book, Page No.90, we can find emissivity of CO₂.

From graph, Emissivity of CO₂ = 0.09

$$\boxed{\varepsilon_{\text{CO}_2} = 0.09}$$

To find correction factor for CO₂

Total pressure, P = 2 atm

$$P_{\text{CO}_2} L_m = 0.06 \text{ m - atm.}$$

From HMT data book, Page No.91, we can find correction factor for CO₂

From graph, correction factor for CO₂ is 1.25

$$\boxed{C_{\text{CO}_2} = 1.25}$$

$$\varepsilon_{\text{CO}_2} \times C_{\text{CO}_2} = 0.09 \times 1.25$$

$$\boxed{\varepsilon_{\text{CO}_2} \times C_{\text{CO}_2} = 0.1125}$$

To find emissivity of H₂O :

$$P_{\text{H}_2\text{O}} \times L_m = 0.1 \times 0.3$$

$$\boxed{P_{\text{H}_2\text{O}} L_m = 0.03 \text{ m - atm}}$$

From HMT data book, Page No.92, we can find emissivity of H₂O.

From graph Emissivity of H₂O = 0.048

$$\boxed{\varepsilon_{\text{H}_2\text{O}} = 0.048}$$

To find correction factor for H₂O:

$$\frac{P_{\text{H}_2\text{O}} + P}{2} = \frac{0.1 + 2}{2} = 1.05$$

$$\frac{P_{\text{H}_2\text{O}} + P}{2} = 1.05,$$

$$P_{\text{H}_2\text{O}} L_m = 0.03 \text{ m - atm}$$

From HMT data book, Page No.92 we can find emission of H₂O

From graph,

Correction factor for H₂O = 1.39

$$C_{\text{H}_2\text{O}} = 1.39$$

$$\varepsilon_{\text{H}_2\text{O}} \times C_{\text{H}_2\text{O}} = 0.048 \times 1.39$$

$$\varepsilon_{\text{H}_2\text{O}} \times C_{\text{H}_2\text{O}} = 0.066$$

Correction factor for mixture of CO₂ and H₂O:

$$\frac{P_{H_2O}}{P_{H_2O} + P_{CO_2}} = \frac{0.1}{0.1 + 0.2} = 1.05$$

$$\boxed{\frac{P_{H_2O}}{P_{H_2O} + P_{CO_2}} = 0.333}$$

$$P_{CO_2} \times L_m + P_{H_2O} \times L_m = 0.06 + 0.03$$

$$\boxed{P_{CO_2} \times L_m + P_{H_2O} \times L_m = 0.09}$$

From HMT data book, Page No.95, we can find correction factor for mixture of CO_2 and H_2O .

From graph, $\Delta\epsilon = 0.002$

$$\begin{aligned} \text{Total emissivity of gaseous mixture, } \epsilon_{\text{mix}} &= \epsilon_{CO_2} \times C_{CO_2} + \epsilon_{H_2O} \times C_{H_2O} - \Delta\epsilon \\ &= 0.1125 + 0.066 - 0.002 \end{aligned}$$

$$\epsilon_{\text{mix}} = \mathbf{0.1765}$$